GURDHARAM JEET SINGH

Full-Stack Software Engineer • Web Systems & Distributed Architecture

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TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, Go, SQL, HTML5, CSS3

Frontend & UI Systems: React 19, Next.js 15 (App Router, Server Components), Tailwind CSS 4, Vite 6, WebSockets, Three.js, Core Web Vitals (LCP, CLS < 0.02)

Backend & Distributed Systems: Node.js, Express, FastAPI, PostgreSQL, Supabase, MariaDB/SQLite Failover, Redis & BullMQ, WebSockets, REST APIs, Docker, Linux/Bash

Cloud & Platform Infrastructure: Official Meta WhatsApp Cloud API (WABA), GitHub Actions 24/7 CI/CD Fleets, Railway, AWS (S3, Lambda), Exotel Telephony, Gemini AI

KEY PRODUCTION PLATFORMS & SYSTEMS

EduCore — Enterprise College Management ERP **2025 – Present**

Lead Full-Stack Architect • React 19, TypeScript, Node.js, Express, MariaDB, Railway Open Source

- **Full-Stack Architecture & Two-Tier Failover:** Engineered enterprise ERP handling student admissions, fee ledgers, and scholarships with automated zero-config DB failover (MariaDB production → local SQLite).
- **RBAC & Dynamic Form Engine:** Built strict JWT role-based access control across 3 permission tiers, Helmet security headers, Zod validation, and a runtime dynamic form schema builder for campus workflows.

Enterprise WhatsApp AI & Lead Management CRM **2025 – Present**

Lead Architect • Meta Cloud API, Node.js, Express, Supabase (PostgreSQL) *Production Platform*

- **Official Meta Cloud API Architecture:** Engineered multi-tenant CRM gateway integrating Meta Graph Cloud API (WABA ID: 106690...), handling bi-directional messaging, template dispatch, and HMAC-verified webhooks.
- **24-Hour Autonomous Follow-Up Radar:** Built real-time engagement engine with dynamic cadences (24h/36h/48h/72h), tracking lead conversation states and triggering automated follow-ups via Supabase.

Nexus Real-Time Quant & Arbitrage Engine **2025 – Present**

Systems Engineer • Python Async FastAPI, WebSockets, Hyperliquid, Binance Futures *High-Throughput Trading Engine*

- **Delta-Neutral Execution Pipeline:** Engineered high-frequency funding rate arbitrage engine analyzing real-time spreads across perpetual futures with automated basis risk calculations and sub-second execution.
- **Real-Time WebSocket Terminal & Kill-Switch:** Built live trading dashboard streaming orderbook feeds over WebSockets with automated 5% drawdown circuit breaker and basis drift emergency unwinds.

SOLUM Minerals & Commercial Web Suite **2025 – Present**

Full-Stack Web Architect • Next.js 15, TypeScript, Tailwind CSS *Commercial Platforms*

- **SOLUM Minerals Geology Portal:** Architected responsive commercial web platform using Next.js 15 App Router, achieving a 99+ Lighthouse performance rating with interactive product showcases and zero layout shift (CLS < 0.02).
- **Branders Digital Experience Platform:** Engineered agency web platform with modern micro-interactions, responsive navigation, and physics-hardened layouts designed for high-conversion client acquisition.

WORK EXPERIENCE

Gurdharam Software Consultancy • Systems & Web Engineer **2025 – Present**

Independent Software Consultancy • Remote *Punjab, India*

- **Telephonic Voice AI Architecture:** Built real-time telephony agent over Exotel WebSockets, SIP, and VoIP, achieving <700ms turn latency across Sarvam AI Indic LLMs, Groq LPU, and Silero VAD barge-in.
- **High-Throughput Webhook Processing:** Architected distributed queue worker system using Redis and BullMQ, reliably processing 25,000+ daily webhook events with automated retry logic and sub-5-minute SLA.

EDUCATION & CREDENTIALS

Baba Farid Group of Institutions (BFGI) — B.Sc. in Computer Science **Expected May 2027**

Core: Data Structures & Algorithms, Database Systems, Computer Networks, Operating Systems *Punjab, India*

Credentials & Verification: Meta Verified Business Entity • Official WhatsApp Business Platform Developer • Sarvam AI Startup Partner • Govt. MSME Registered Enterprise (UDYAM-PB-03-0079744).